

DR ESLAM AHMED
eliteigcse.com

CHAPTER 1 • NUMBER • TOPIC 12

Ratio Toolkit

A ratio is parts. Find the value of one part, then build every answer from it — scale, share, link and solve.



$$\frac{3}{10} = 0.3 =$$

30%**5**

VARIANTS

5

PAST-PAPER PARTS

55

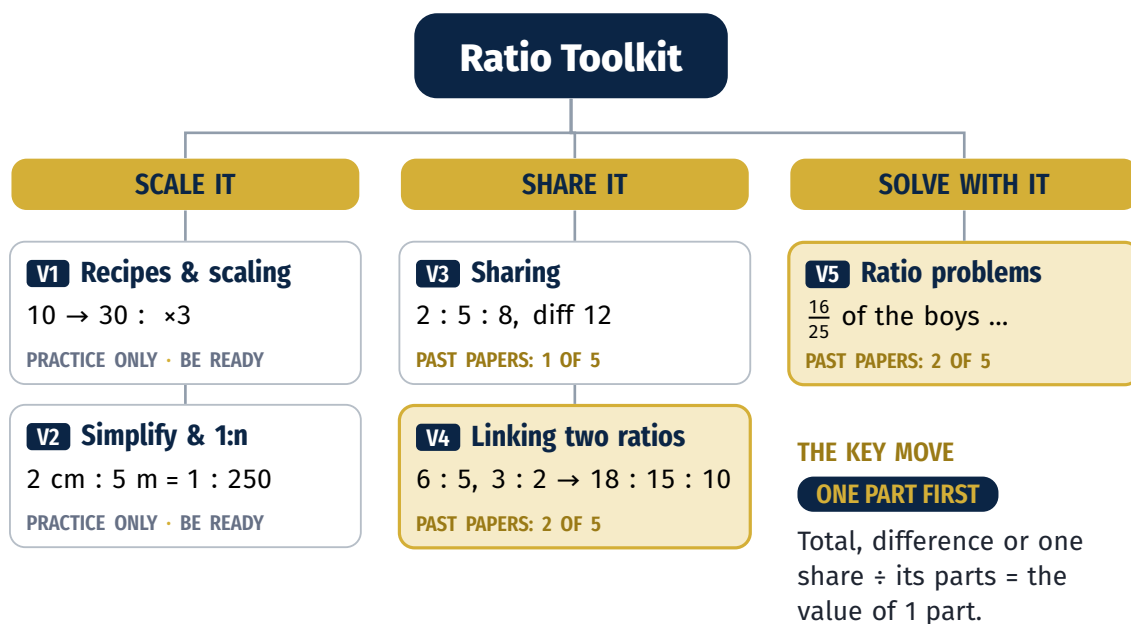
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Recipes & scaling**V2** Simplify & 1:n**V3** Sharing**V4** Linking two ratios**V5** Ratio problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Ratio at a glance



Your notes


★ How often does each variant come up?

5 past-paper questions (Jan 2020 → Nov 2023); 55 practice parts — grouped by type. More ratio questions are in Topic 13.

	PAST PAPERS (OUT OF 5)		PRACTICE PARTS	
V1 Recipes, scaling	● ● ● ● ●	0		13
V2 Simplify and 1:n	● ● ● ● ●	0		8
V3 Sharing	● ● ● ● ●	1		15
V4 Linking two ratios	● ● ● ● ●	2		2
V5 Ratio problems	● ● ● ● ●	2		19

WHAT THIS TELLS YOU The exam skips the easy scaling and goes straight to linking two ratios and multi-step stories (3–6 marks, early in Paper 1 or 2). Every one is cracked by finding one part first.

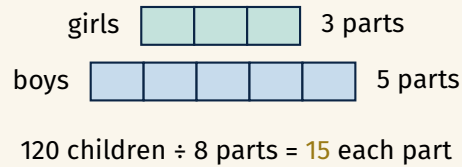
BEFORE YOU START

The big idea

**A ratio is a number of equal parts.**

Girls : boys = 3 : 5 means for every 3 girls there are 5 boys — **8 parts** in total.

Once you know what **one part** is worth, everything else is multiplying.


 The four numbers to find
Parts

$$3 : 5$$

Each number is a count of equal parts.

Total

$$3 + 5 = 8$$

Add the parts when a total is given.

Difference

$$5 - 3 = 2$$

Subtract the parts when “more than” is given.

One part

$$120 \div 8 = 15$$

Divide what you know by its parts.

 Which number do you divide by?
“share £120”

$$120 \div (3 + 5)$$

A total: divide by the total parts.

“12 more than”

$$12 \div (5 - 2)$$

A difference: divide by the difference.

“120 g of sugar”

$$120 \div 2$$

One share: divide by its own parts.

**CALCULATOR FIRST, THEN CHECK**

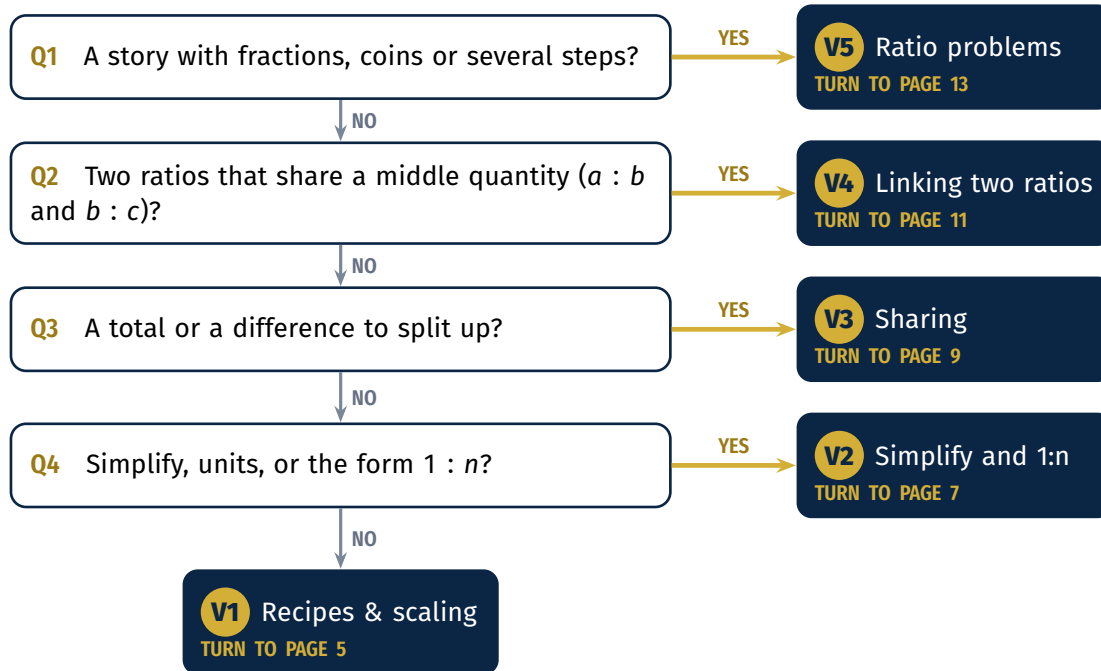
Check the answer: the shares must add to the total and still be in the given ratio (45 : 75 = 3 : 5 ✓).

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Recipes & Scaling

Find the multiplier (new $\div$ old) and use it on every amount.

SEEN IN PAST PAPERS

● ● ● ● ● 0/5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 13


 Your notes

? What is it?

In plain words

A recipe makes a fixed number. To make more or fewer, find the **multiplier**: new $\div$ old.

Multiply **every ingredient** by it.
 “Greatest number you can make”: find each ingredient’s multiplier — the **smallest** one wins.

IT LOOKS LIKE

10 pancakes $\rightarrow$ 30

enough flour?

greatest number

10 pancakes
120 g flour

$\times 3$ (30 $\div$ 10)

30 pancakes
360 g flour

How do I do it?

1 Multiplier

new $\div$ old (people, cakes, biscuits).

2 Scale

Every ingredient $\times$ the multiplier.

3 Limit

Greatest number: have $\div$ need for each — take the smallest.

4 Answer

Units; “enough?” — compare and say Yes/No.

NEW $\div$ OLD = MULTIPLIER Same multiplier for every ingredient.

V1 Recipes & Scaling

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Ingredients for 4 people ... for 6 people"*→ **Multiplier** $6 \div 4 = 1.5$ *"Does she have enough?"*→ **Work out what she needs, compare, then Yes/No***"The greatest number he can make"*→ **Have ÷ need for each; smallest wins****WORKED EXAMPLE** **PRACTICE**

Practice • Easy • Q1 • 4 marks

10 pancakes need 300 ml milk, 120 g flour, 2 eggs. (a) How much flour for 30 pancakes? (b) Tara uses 750 ml of milk. How many pancakes?

- | | | |
|---------------|---------------------------------------|--------------------------------|
| STEP 1 | $30 \div 10 = 3$ | ◀ (a) multiplier |
| STEP 2 | $120 \times 3 = 360 \text{ g}$ | ◀ scale the flour |
| STEP 3 | $750 \div 300 = 2.5$ | ◀ (b) multiplier from the milk |
| STEP 4 | $10 \times 2.5 = 25 \text{ pancakes}$ | ◀ scale the pancakes |

ANSWER (a) 360 g (b) 25 pancakes ✓ 25 pancakes need 750 ml ✓

WATCH OUT✗ $120 + 20 = 140 \text{ g}$ ✓ $120 \times 3 = 360 \text{ g}$

Recipes scale by multiplying, never by adding.

EXAM TIP

"Enough?" is a show question: write what is needed, what she has, then Yes or No.

Check yourself

calculator first • answers on page 15

1.1 Practice • Easy

16 gingerbread men:
180 g flour, 40 g ginger,
110 g butter, 30 g sugar.
Amounts for 24?

Answer:

1.2 Practice • Easy

225 g flour makes 9
cakes. Is 475 g enough
for 20 cakes?

Answer:

1.3 Practice • Medium

8 biscuits: 120 g butter,
60 g sugar, 180 g flour.
He has 330 g, 200 g, 450
g. Greatest number?

Answer:



VARIANT 2

VARIANT 2

Simplify & the Form

1:n

Same units, divide by the HCF — or divide by the first number for 1:n.

SEEN IN PAST PAPERS

0 / 5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 8



Your notes

❓ What is it?

In plain words

Simplify: divide every part by the same number (the HCF), like a fraction.

Units first: 2 cm : 5 m must become 2 : 500. Form **1 : n**: divide both parts by the first number.

IT LOOKS LIKE

$$24 : 36 : 60$$
$$50 \text{ g} : 1 \text{ kg} = 1 : n$$
$$2 \text{ cm} : 5 \text{ m}$$

SAME UNITS

2 : 500

÷2 BOTH

1 : 250

How do I do it?

1 Units

Change to the smaller unit: m $\rightarrow$ cm, kg $\rightarrow$ g.

2 Divide

Every part by the HCF (simplest form).

3 1:n

Divide every part by the first number.

4 Check

Whole numbers for “simplest”; n may be a decimal.

SAME UNITS FIRST A ratio has no units once simplified.

V2 Simplify & the Form 1:n

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Write in its simplest form"*→ **Divide by the HCF until nothing divides all parts***"In the form 1 : n"*→ **Divide everything by the first number***Mixed units: cm and m, g and kg*→ **Convert first****WORKED EXAMPLE** **PRACTICE**

Practice · Easy · Q10(a) · 2 marks

The ratio 2 centimetres to 5 metres can be written in the form $1 : n$. Find the value of n .

STEP 1 5 m = 500 cm

◀ same units

STEP 2 2 : 500

◀ the ratio

STEP 3 1 : 250◀ $\div 2$ both**ANSWER** $n = 250$ ✓ $2 \times 250 = 500$ ✓**WATCH OUT** $\times$ 2 cm : 5 m = 2 : 5

✓ 2 : 500 = 1 : 250

Different units make the ratio wrong by a factor of 100.

EXAM TIP

$1 : n$ is ideal for maps and scales — Topic 13 and scale drawings use it.

Check yourself

calculator first · answers on page 15

2.1 Practice · Easy

The ratio 50 grams to 1 kilogram can be written as $1 : n$. Find n .

Answer:

2.2 Practice · Easy

A is half of B. Work out the ratio A : B.

Answer:

2.3 Dr Eslam

Simplify 24 : 36 : 60.

Answer:

V3

VARIANT 3

Sharing: the Value of One Part

Divide the total (or the difference) by its number of parts, then multiply.

SEEN IN PAST PAPERS

● ● ● ● ● 1 / 5

PRACTICE QUESTIONS: 15


Your notes

Came up in: May 2023 P1H Q1

? What is it?

In plain words

Given a **total**: $\text{total} \div (\text{sum of the parts}) = \text{one part}$.

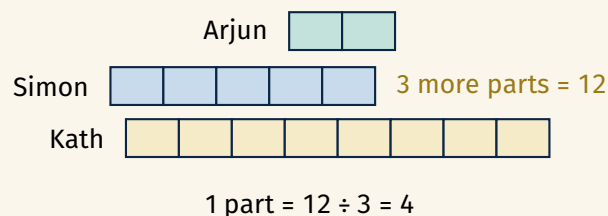
Given a **difference** ("12 more than"): $\text{difference} \div (\text{difference of the parts}) = \text{one part}$.

Then each share = $\text{its parts} \times \text{one part}$.

IT LOOKS LIKE

share £630 in 5 : 13

12 more goals



How do I do it?

1 Parts

Write the ratio and count the parts.

2 Match

Total ↔ sum of parts; difference ↔ difference of parts.

3 One part

Divide.

4 Shares

Multiply each number in the ratio by one part.

ONE PART FIRST Then every share is a multiplication.

V3 Sharing: the Value of One Part

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Share £630 in the ratio 5 : 13" → $630 \div 18 = 35$ **per part**

"Simon scored 12 more goals than Arjun" → $12 \div (5 - 2)$ **per part**

"How much does Harry receive?" → **Only his share: his parts × one part**

WORKED EXAMPLE **PAST PAPER**

May 2023 • Paper 1H • Q1 • 3 marks

Goals scored by Arjun, Simon and Kath were in the ratios 2 : 5 : 8. Simon scored 12 more goals than Arjun. Work out the number of goals scored by Kath.

STEP 1 2 : 5 : 8 ◀ parts

STEP 2 $5 - 2 = 3$ parts = 12 ◀ difference ↔ difference

STEP 3 1 part = $12 \div 3 = 4$ ◀ one part

STEP 4 Kath = $8 \times 4 = 32$ ◀ her share

ANSWER 32 goals✓ Arjun 8, Simon 20: $20 - 8 = 12$ ✓**WATCH OUT**✗ $12 \div 15$ per part✓ $12 \div (5 - 2) = 4$

12 is a difference, so divide by the difference in parts, not the total.

EXAM TIP

"Write down all the stages": show the 3 parts, the 4 per part, then 8×4 .

Check yourself

calculator first • answers on page 15

3.1 Practice • Easy

153 cm of string cut in the ratio 4 : 2 : 3. Each length?

Answer:

3.2 Practice • Easy

Share £630 between Daniel and Rose in the ratio 5 : 13.

Answer:

3.3 Dr Eslam

Boys : girls = 5 : 3.
There are 18 more boys than girls. How many students?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Linking Two Ratios

Make the shared quantity the same number in both ratios, then join them.

SEEN IN PAST PAPERS

●●●●● **2/5**

PRACTICE QUESTIONS: 2



Your notes

Came up in: May 2021 P1H Q4 • Nov 2023 P2H Q5

? What is it?

In plain words

Two ratios share a

middle quantity: flour : butter
= 6 : 5 and butter : sugar = 3 : 2.

Butter is 5 in one and 3 in the other. Scale both so butter is the **LCM**, 15. Then join: 18 : 15 : 10.

IT LOOKS LIKE

$$a : b = 3 : 4$$

$$b : c = 6 : 5 \Rightarrow a : b : c$$

$$6 : 5 \quad (\times 3) \rightarrow 18 : 15$$

$$3 : 2 \quad (\times 5) \rightarrow 15 : 10$$

$$18 : 15 : 10$$

MATCH THE MIDDLE: LCM OF 5 AND 3

How do I do it?

1 Middle

Find the quantity in both ratios.

2 LCM

Make it the same number (LCM) in both.

3 Join

Write one three-part ratio.

4 One part

Use the given amount $\div$ its parts; multiply.

MATCH THE MIDDLE 2 of the 5 past papers.

V4 Linking Two Ratios

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"chickens : ducks = 7 : 2, ducks : pigs = 5 : 9" → **Ducks to 10:** 35 : 10 : 18

"She uses 120 grams of sugar" → 120 ÷ **sugar's parts** = **one part**

"Write $a : b : c$ in its simplest form" → **Match the middle, then simplify**

 WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q5 · 3 marks

flour : butter = 6 : 5 and butter : sugar = 3 : 2. Avril uses 120 grams of sugar. Work out the weight of flour she uses.

STEP 1 6 : 5 ($\times 3$) = 18 : 15, 3 : 2 ($\times 5$) = 15 : 10 ◀ butter to 15

STEP 3 flour : butter : sugar = 18 : 15 : 10 ◀ join

STEP 4 1 part = $120 \div 10 = 12$, $18 \times 12 = 216$ ◀ one part, then flour

ANSWER 216 g

✓ 216 : 180 : 120 = 6 : 5 and 3 : 2 ✓

 WATCH OUT

✗ 6 : 5 : 2

✓ 18 : 15 : 10

Butter can't be 5 and 3 at once — match it first.

 EXAM TIP

Check: divide your joined ratio back to both given ratios.

 Check yourself

calculator first · answers on page 15

4.1

May 2021 P1H

chickens : ducks = 7 : 2,
ducks : pigs = 5 : 9.
There are 36 pigs. How
many chickens?

Answer:

4.2

Dr Eslam

$a : b = 3 : 4$ and
 $b : c = 6 : 5$. Write
 $a : b : c$ in its simplest
form.

Answer:

4.3

Dr Eslam

$x : y = 2 : 3$ and
 $y : z = 4 : 5$. Find
 $x : y : z$.

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM**Ratio Problems**

Split the story into steps; each step is a ratio, a fraction or a one-part calculation.

SEEN IN PAST PAPERS

●●●●● **2 / 5**

PRACTICE QUESTIONS: 19



Your notes

Came up in: Jan 2020 P1H Q5 • Jun 2023 P2H Q5

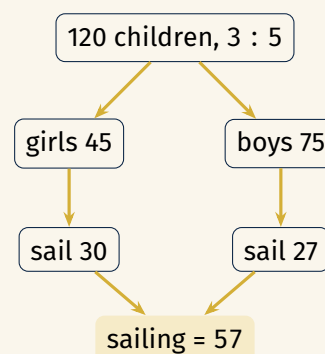
? What is it?**In plain words**

A longer story mixes ratio with **fractions**, **money values** or a **change**. Do it one step at a time and label each number.

With coins, a **group** is one set of the ratio: 1 : 3 means one 2p and three 5p = 17p per group.

IT LOOKS LIKE $\frac{16}{25}$ of the boys

85p in 2p : 5p = 1 : 3

**How do I do it?****1 Split**

Use the first ratio to get the groups.

2 Step on

Apply each fraction or ratio to the right group.

3 Label

Write what each number means (girls sail, boys sail ...).

4 Answer

Combine what the question asks for.

ONE STEP, ONE LABEL 5–6 marks: method marks for each step.

V5 Ratio Problems

VARIANT 5 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*" $\frac{16}{25}$ of the boys go climbing"*→ **Fraction of that group only***Coins with a value and a ratio*→ **Value of one group; total ÷ group****WORKED EXAMPLE** **PAST PAPER**

January 2020 • Paper 1H • Q5 • 6 marks

120 children; girls : boys = 3 : 5. $\frac{16}{25}$ of the boys go climbing. Twice as many girls go sailing as go climbing. Everyone either sails or climbs. How many children go sailing?

STEP 1 $120 \div 8 = 15$: girls 45, boys 75 ◀ split 3 : 5

STEP 2 boys climb $\frac{16}{25} \times 75 = 48$, sail 27 ◀ fraction of the boys

STEP 3+4 girls sail $\frac{2}{3} \times 45 = 30$; $27 + 30 = 57$ ◀ sail : climb = 2 : 1; add
ANSWER 57 children✓ climb $48 + 15 = 63$; $57 + 63 = 120$ ✓**WATCH OUT**

✗ $\frac{16}{25} \times 120$

✓ $\frac{16}{25} \times 75$

The fraction is of the boys, not of everyone.

EXAM TIP

Check the end: sailing + climbing must give back 120.

Check yourself

calculator first • answers on page 15

5.1 Jun 2023 P2H

85p in 2p and 5p coins;
 $2p : 5p = 1 : 3$. How
 many more 5p coins?

Answer:

5.2 Dr Eslam

Red : blue = 2 : 3. Add
 8 red: now 1 : 1. How
 many blue?

Answer:

5.3 Dr Eslam

Ages 2 : 3 now, 3 : 4 in
 5 years. Ages now?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 270, 60, 165, 45 g**1.2** 500 g needed: No**1.3** 20 biscuits**2.1** $n = 20$ **2.2** 1 : 2**2.3** 2 : 3 : 5**3.1** 68, 34, 51 cm**3.2** Daniel £175, Rose
£455**3.3** 72**4.1** 70 chickens**4.2** 9 : 12 : 10**4.3** 8 : 12 : 15**5.1** 10 more**5.2** 24 blue**5.3** 10 and 15

THE WHOLE TOPIC ON ONE PAGE

Ratio cheat sheet



Your notes
V1 Recipes & scaling

$$\times \frac{\text{new}}{\text{old}}$$

Same multiplier for every amount.

V2 Simplify & 1:n

$$2 : 500 = 1 : 250$$

Same units; divide all parts.

V3 Sharing 1/5

$$12 \div (5 - 2) = 4$$

Total $\leftrightarrow$ sum; difference $\leftrightarrow$ difference.**V4 Linking 2/5**

$$18 : 15 : 10$$

Match the middle (LCM), join.

V5 Problems 2/5

$$45, 75 \rightarrow 30 + 27 = 57$$

One step, one label.

**TOP 3 LOST MARKS**

1. dividing a difference by the total $\times$
2. joining ratios without matching $\times$
3. mixed units $\times$

BOTTOM LINE Ratio = parts • one part first • match the middle • one step, one label.